EVERY FIELD HAS AN ALGEBRAIC CLOSURE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The author interjects before the proof (source lines 1188-1191) that this existence result is mostly a red herring for the rest of the chapter; the interjection is outside the counted span and is not delivered.

Editorial note. The label theorem-existence-algebraic-closure names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own theorem environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 1169-1177: definition of algebraic closure (algebraic + algebraically closed)
- Same chapter: lemma-size-algebraic-extension (cardinality of an algebraic extension of F is at most max(aleph_0, |F|)), lemma-algebraically-closed (equivalent characterizations, so not algebraically closed means an irreducible polynomial of degree > 1 exists), lemma-finite-is-algebraic and lemma-algebraic-permanence (algebraicity is stable in towers)

Source reference. Original reference 09GJ: sources/stacks-fields/pinned-original-source.tex, source lines 957–957. Prerequisite pointer invoked at 1231; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09GR: sources/stacks-fields/pinned-original-source.tex, source lines 1128–1128. Prerequisite pointer invoked at 1228; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09GG: sources/stacks-fields/pinned-original-source.tex, source lines 878–878. Prerequisite pointer invoked at 1229; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09GK: sources/stacks-fields/pinned-original-source.tex, source lines 986–986. Prerequisite pointer invoked at 1194; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Definition 0.1. Let F be a field. An *algebraic closure* of F is a field $\overline{F}$ containing F such that:

(1) $\overline{F}$ is algebraic over F.

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) $\overline{F}$ is algebraically closed.

Theorem 0.2. *Every field has an algebraic closure.*

Proof. Let F be a field. By Lemma 09GK the cardinality of an algebraic extension of F is bounded by $\max(\aleph_0, |F|)$. Choose a set S containing F with $|S| > \max(\aleph_0, |F|)$. Let's consider triples (E, σ_E, μ_E) where

(1) E is a set with $F \subset E \subset S$, and
(2) $\sigma_E : E \times E \to E$ and $\mu_E : E \times E \to E$ are maps of sets such that (E, σ_E, μ_E) defines the structure of a field extension of F (in particular $\sigma_E(a, b) = a +_F b$ for $a, b \in F$ and similarly for μ_E), and
(3) E/F is an algebraic field extension.

The collection of all triples (E, σ_E, μ_E) forms a set I. For $i \in I$ we will denote $E_i = (E_i, \sigma_i, \mu_i)$ the corresponding field extension to F. We define a partial ordering on I by declaring $i \leq i'$ if and only if $E_i \subset E_{i'}$ (this makes sense as E_i and $E_{i'}$ are subsets of the same set S) and we have $\sigma_i = \sigma_{i'}|_{E_i \times E_i}$ and $\mu_i = \mu_{i'}|_{E_i \times E_i}$, in other words, $E_{i'}$ is a field extension of E_i.

Let $T \subset I$ be a totally ordered subset. Then it is clear that $E_T = \bigcup_{i \in T} E_i$ with induced maps $\sigma_T = \bigcup \sigma_i$ and $\mu_T = \bigcup \mu_i$ is another element of I. In other words every totally order subset of I has a upper bound in I. By Zorn's lemma there exists a maximal element (E, σ_E, μ_E) in I. We claim that E is an algebraic closure. Since by definition of I the extension E/F is algebraic, it suffices to show that E is algebraically closed.

To see this we argue by contradiction. Namely, suppose that E is not algebraically closed. Then there exists an irreducible polynomial P over E of degree > 1, see Lemma 09GR. By Lemma 09GG we obtain a nontrivial finite extension $E' = E[x]/(P)$. Observe that E'/F is algebraic by Lemma 09GJ. Thus the cardinality of E' is $\leq \max(\aleph_0, |F|)$. By elementary set theory we can extend the given injection $E \subset S$ to an injection $E' \to S$. In other words, we may think of E' as an element of our set I contradicting the maximality of E. This contradiction completes the proof. $\square$